



One-Footed Bride Tuning

Extending Bach's Emotional Power Through Just Intonation

Prent Rodgers

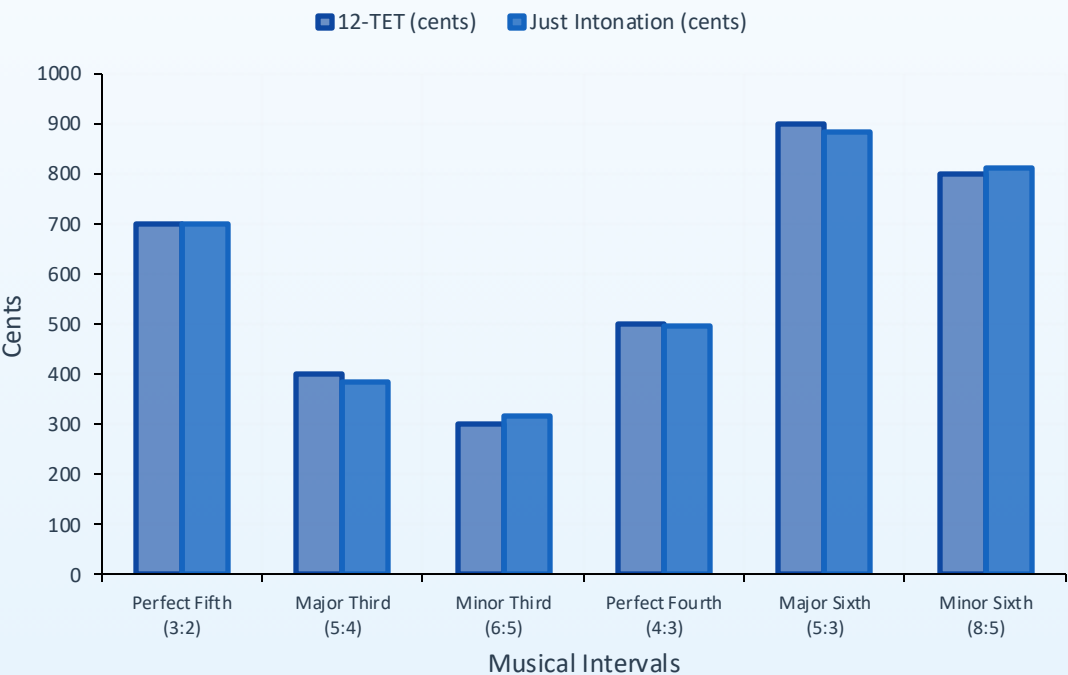
<https://github.com/prentrogers/One-footed-bride-tuning>

2/14/26

Computational Optimization of Harry Partch's Microtonal System
Applied to Bach Wedding Chorales

The Challenge: Equal Temperament vs Just Intonation

Interval Comparison: 12-TET vs Just Intonation



Why Just Intonation Matters

12-TET Compromise: All semitones equal **100 cents** , but intervals are mathematically impure for harmonic resonance.

Just Intonation Purity: Intervals based on small integer ratios like **3:2** (perfect fifth) and **5:4** (major third).

Emotional Impact: Pure ratios create stronger harmonic resonance, enhancing the emotional power of Bach's four-part harmonies.

Computational Solution: Simulated annealing finds optimal rational intervals while preserving Bach's harmonic structure.

***Key Insight:** By replacing 12-TET's approximations with pure rational intervals, we unlock the natural harmonic relationships that amplify musical emotion.*

Harry Partch's 11-limit Tonality Diamond

Mathematical Foundation

 **Diamond Structure** organizes rational intervals within prime limits. Overtones go up and undertones go down

 **Score** = numerator + denominator, $\frac{3}{2} = 5$.

 **31-limit** contains 214 unique intervals.

Key Insight: Complete palette of just intervals for microtonal harmony optimization.

						2/1						
					7/4		16/9					
				3/2		14/9		8/5				
			11/8		4/3		7/5		16/11			
		5/4		11/9		6/5		14/11		8/3		
	9/8		10/9		11/10		12/11		7/3		16/7	
1/1		1/1		1/1		1/1		1/1		1/1		2/1
	16/9		9/10		10/11		11/6		12/7		7/4	
		8/5		9/11		5/3		11/7		3/2		
			16/11		3/2		10/7		11/8			
				4/3		9/7		5/4				
					8/7		9/8					
						1/1						

28 unique pitches

The One-Footed Bride Concept

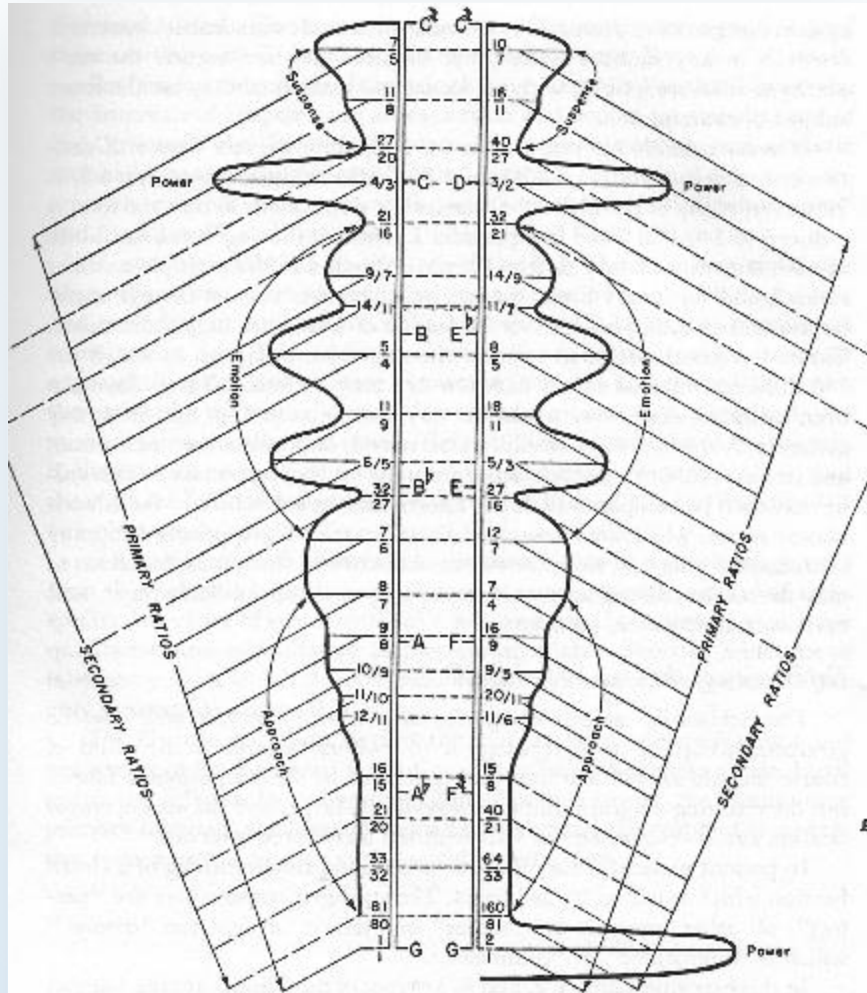


DIAGRAM 8.—THE ONE-FOOTED BRIDE: A GRAPH OF COMPARATIVE CONSONANCE

Partch's Consonance Theory

- ◆ **Tonality Diamond** organizes rational intervals by **limit complexity** and consonance relationships.
- ♥ **Intervals of Emotion** use small ratios like **5/4, 6/5, 7/6** for expressive harmonic content.
- ⚡ **Intervals of Power** employ **3/2, 4/3** ratios for structural harmonic foundation.
- 📊 **Consonance Score** equals numerator plus denominator, where lower scores indicate greater consonance.

Core Innovation: The One-Footed Bride diagram maps harmonic relationships spatially, revealing how rational intervals create natural consonance hierarchies that our optimization algorithm exploits.

Simulated Annealing Optimization

Algorithm Parameters

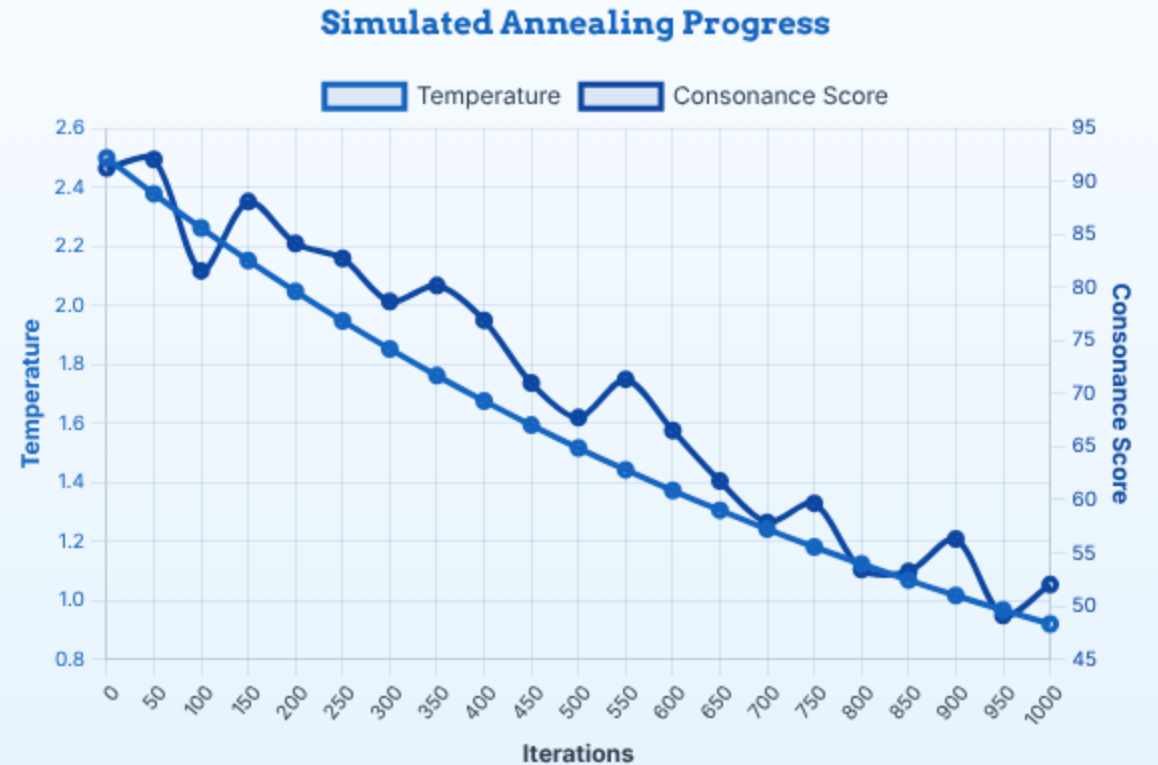
Initial Temperature: 2.5 - High exploration phase allows wide search of ratio space.

Cooling Rate: 0.999 per iteration - Gradual transition from exploration to exploitation.

Tolerance: 1 cent - Precision threshold for interval matching within the tonality diamond.

Multi-Roll Strategy: 5-8 rolls per chord to escape local minima and find global optimum.

Core Innovation: The algorithm systematically explores Partch's tonality diamond to find 4-note chord tunings that minimize consonance scores while preserving Bach's harmonic structure.



Bach Chorale Analysis Pipeline

Complete Workflow: MIDI to Optimized Audio



MIDI Input

Load Bach Wedding Chorales **BWV 253-264** using **music21** library



Diamond Generation

Build **n-limit tonality diamond** with variable number of rational intervals



Simulated Annealing

Optimize 4-note chords using **temperature scheduling** and multi-roll strategy



Caching System

Achieve **~80% cache hit rate** through chord compression



Glissando Smoothing

Add **voice leading continuity** for natural transitions



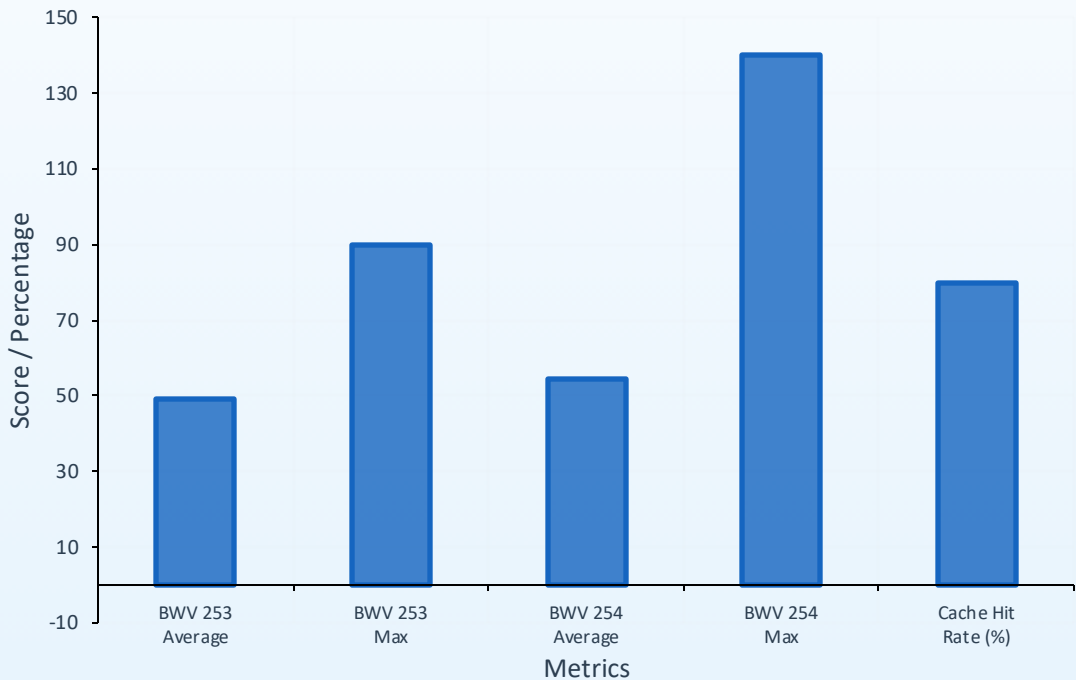
Csound Synthesis

Render **24-bit WAV** files using sample-based instruments

Technical Achievement: Automated pipeline processes complete Bach chorales from MIDI input to optimized just intonation audio in minutes, preserving harmonic structure while enhancing emotional resonance through pure rational intervals.

Results: Consonance Scores & Performance

Bach Chorale Optimization Results



Optimization Results

BWV 253 Average Score: 48.9 with maximum chord score of 90.0, demonstrating effective consonance optimization.

BWV 254 Average Score: 54.6 with maximum chord score of 140.0, showing varied harmonic complexity across chorales.

Cache Performance: ~80% hit rate from chord compression, dramatically accelerating repeated optimizations.

Tolerance Levels: Configurable from 1-4 cents, enabling fine-tuned control over harmonic precision.

***Key Achievement:** Systematic reduction of consonance scores while preserving Bach's harmonic structure, proving computational optimization can enhance musical expression.*

Audio Synthesis with Csound

Sample-Based Synthesis Pipeline

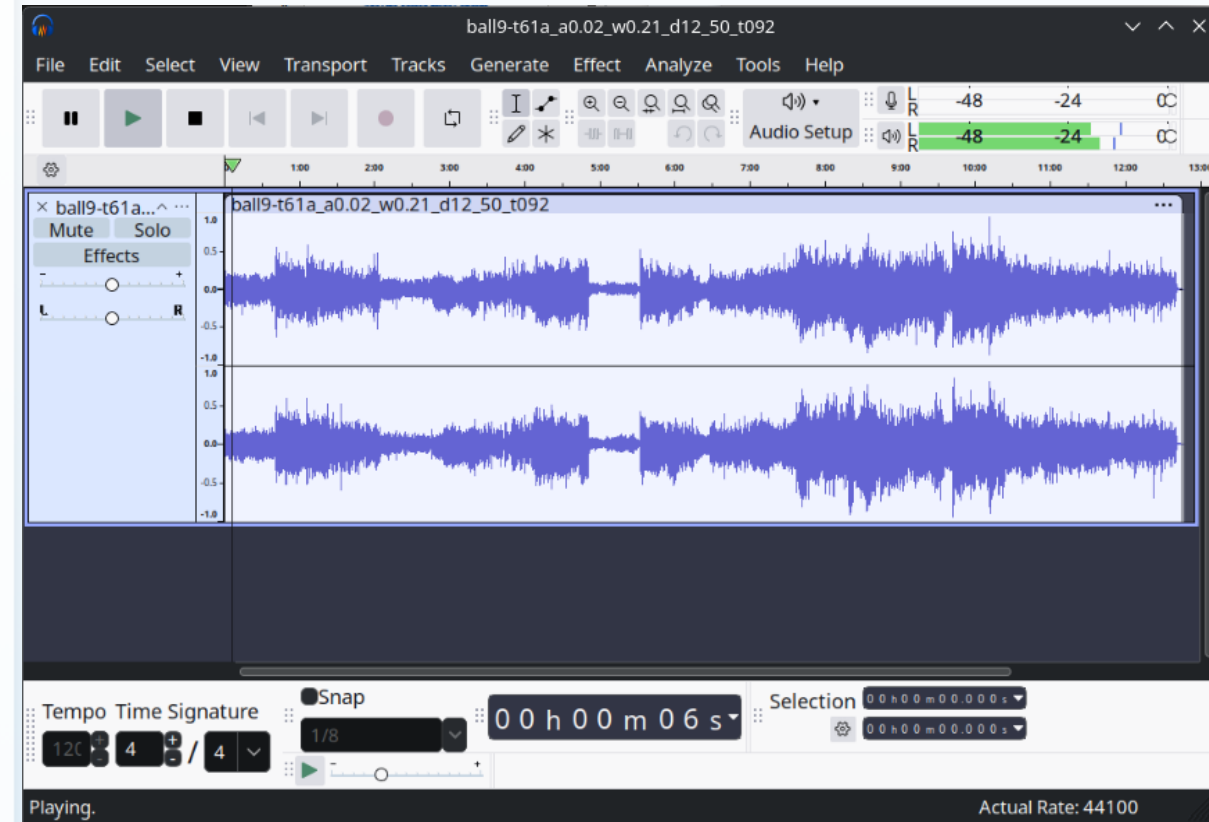
 **Sample Library Integration** uses McGill samples or my own samples for realistic instrument timbres.

 **Precise Cent Detuning** applies just intonation ratios using `diskin2` and pitch shifting opcodes.

Glissando Smoothing generates interpolation tables for natural voice leading between chord transitions.

 **Dynamic Processing** includes reverb, compression, and multi-instrument routing for professional audio quality.

***Technical Achievement:** The Csound pipeline preserves natural instrument character while enabling precise microtonal control impossible with traditional MIDI synthesis.*



Key Innovations & Impact

Technical Contributions

-  **Simulated Annealing** optimization successfully finds low-ratio intervals for complex 4-note chords.
-  **Tonality Diamond Integration** provides systematic access to `19-47 limit` rational intervals.
-  **Bach Chorale Enhancement** demonstrates measurable consonance improvement through just intonation.
-  **Csound Pipeline** enables high-quality sample-based synthesis with precise microtonal control.
-  **Performance Optimization** achieves >80% cache hit rates through intelligent chord compression.

Explore the Project

Experience the **emotional power** of just intonation applied to Bach's masterworks. The complete implementation is available for research and experimentation.

 [View on GitHub](#)

Complete Python implementation with Bach Wedding Chorales (BWV 253-264)